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REVIEW ARTICLE

APPLICATION OF LINEAR PROGRAMMING FOR COST MINIMIZATION OF FISH FEED

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ABSTRACT

The use of the linear programming technique has increased in a number of applications, including the traffic light control, the allocation of aircraft over various routes, the management of water quality in warehouses, caterers, personnel, and advertising media selection problems. It has recently had a significant impact on research in the fields of agriculture, animal husbandry, and livestock and it may be used to determine the components of fish feed for fish farmers in order to increase fish productivity. Fish farmers in Bangladesh's Jashore district feed their fish using a traditional manner, although contemporary fish feed is made to stringent nutrient criteria that are important for fish growth and increased animal productivity. This study discusses the formulation of fish feed using a flexible tool known as the "linear programming technique".

KEYWORDS

Linear Programming Model, Simplex Method, Decision Variables, Optimal Results

1. INTRODUCTION

Linear programming (LP) is a mathematical method for determining the most efficient use of a company's limited resources to accomplish the desired outcome. It is also a mathematical strategy for determining the ideal product combination to maximize profit or reduce cost within the limitations of available resources. The systematic method of figuring out the greatest or least value of any variable in a function using linear programming is referred to as a mathematical problem or an optimization problem. Most of the time, the problem is given as a linear function that needs to be optimized while taking into consideration a variety of specific constraints. It's the system for choosing the best product combination to maximize profit or minimize cost within the limits of the possibility to get resources.

The main application of linear programming problems is in providing management guidance on how to utilize limited resources as effectively and efficiently as possible. Planning, manufacturing, transportation, technology, and other challenges in microeconomics and firm operation depend on linear programming. As a result, numerous problems can be classified as linear programming problems. Utilizing the linear programming technique in several domains, many researchers worked in tiny manufacturing firms to maximize profit or minimize cost.

A study in 2011 discussed with aquaculture species of the freshwater air-breathing African catfish (Clariasgariepinus) (Farhat, 2011). It is important commercially because of its disease resistance, high fertility, and simplicity of production in captivity. Seven locally mobilized components from Uganda's nearby nutrients were investigated by (Kwikiriza et al., 2016. The component's moisture, ash, crude fat, crude protein, gross energy contents, and matter basis were analyzed using approved methods. In 2014, a study described how the development of linear programming and the use of software tools had significantly impacted the composition of fish feed (Nath et al. 2014). By using the conventional linear encoding problem on price minimization on fish feeds, in 2019 researchers developed with the goal of reducing the cost of producing fish feeds (Palani et al., 2019). Both primary and secondary data were used to acquire the data. The data were evaluated using a linear programming problem, and the best solution was found after second iterations using fingerlings. The goal of this research was to reduce the price of producing fish feeds through the use of a linear programming problem to cost reduction on fish feeds. The creation and testing of a demand feeder for carp feeding in an outdoor culture system was discussed by (Mohapatra et al., 2009). The feed conversion ratio (FCR) in the demand-fed system was superior. It can save a lot of cost on feed and labor time when used.

In 1988, a study discussed the analysis of production planning in fish processing enterprises (Jensson et al., 1988). A five-day linear programming model that maximizes the firms' profits is described, and the results of actual testing are examined. In 2021 a study examined in order to manufacture ayurveda mixtures or pastes that increase human immunity and meet vitamin requirements while maximizing profit (Jain et al., 2021). The ayurvedic products production industry uses linear programming to find the best production strategy. Researchers in 2019 invented with linear programming technique for production planning in an engineering industry and this study creates a mathematical model to determine the ideal staff, lot size, and needed capacity (Jyothi et al., 2019). A research examined the economic viability, attractiveness, and sustainability of fish farming because they are readily available, inexpensive, and do not compete with human consumption (Ibrahim et al., 2016).

This paper is organized into four sections. Various journal articles and some archival studies are included in this current section. In the second section, we discuss the methodology, which is used to conduct methodical research. In next section, the acquired data is subjected to a simple test to determine the issues by using the simplex system of linear programming problems. In the last section, we represent the results and discussion. Results of our paper contain in section 4. The conclusion of this paper is

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presented in the final section.

2. Methodology of Linear Programming Problem Formulation

A tabular representation of the general linear programming problems is shown below:

Table 1: General linear programming problems

Ingredients	x_1	x_2	⋯	x_n	Solutions
1	p_{11}	p_{12}	⋯	p_{1n}	q_1
2 ⋮ ⋮	p_{21} ⋮ ⋮	p_{22} ⋮ ⋮	⋯	p_{2n} ⋮ ⋮	q_2 ⋮ ⋮
M	p_{m1}	p_{m2}	⋯	p_{mn}	q_n
Cost (Taka)	c_1	c_2	⋯	c_n	

The following format can be used to interpret the above table:

$$\text{Optimize } Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n \quad (1)$$

Subject to the constraints:

$$\left.\begin{array}{l} p_{11}x_1 + p_{12}x_2 + \cdots + p_{1n}x_n \leq \text{ or } \geq q_1 \\ p_{21}x_1 + p_{22}x_2 + \cdots + p_{2n}x_n \leq \text{ or } \geq q_2 \\ p_{31}x_1 + p_{32}x_2 + \cdots + p_{3n}x_n \leq \text{ or } \geq q_3 \\ \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\ p_{m1}x_1 + p_{m2}x_2 + \cdots + p_{mn}x_n \leq \text{ or } \geq q_m \end{array}\right\} \quad (2)$$

and the non-negativity restrictions are:

$$x_1, x_2, x_3, \ldots, x_n \geq 0 \quad (3)$$

We carry out the following steps in order to solve a common maximizing problem:

1. Adding slack variables and rewriting each inequality as an equation. That is,

$$p_{i1}x_1 + p_{i2}x_2 + \cdots + p_{in}x_n \leq \text{ or } \geq q_i$$

becomes

$$p_{i1}x_1 + p_{i2}x_2 + \cdots + p_{in}x_n + s_i = q_i$$

where $i = 1,2,3, \ldots\ldots\ldots\ldots, m$ and s_i is slack variable.

2. The objective function is rewritten as follows:

$$Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n = 0$$

3. Making an array with the coefficients of our variables and slack variables for the simplex table and P.

Table 2: An array with the coefficients of our variables and slack variables

x_1	x_2	……	x_n	s_1	s_2	……	s_m	Z	Constant
p_{11}	p_{12}	……	p_{1n}	1	0	……	0	0	q_1
.	.	……	.	0	1	……	0	0	.
.	.	……	.	0	0	……	0	0	.
.	.	……	.	0	0	……	0	0	.
p_{m1}	p_{m2}	……	p_{mn}	0	0	……	1	0	q_m
$-c_1$	$-c_2$	……	$-c_n$	0	0	……	0	1	0

4. Verifying whether the optimal solution has been identified by going over each item in the final row to the left of the vertical line. The optimum solution has been identified if every single entry is nonnegative. Next, we go to Step 7. It has not been possible to find the optimal solution if there are one or more negative entries. And now we go on to Step 5.

5. To find the pivot column, find the largest negative value in the last row to the left of the vertical line. Next, divide each element in the constant column by the pivot column's corresponding entry. The row that represents the pivot is the ratio with the lowest value; if any of the quotients are negative or have no meaning, we disregard this row. If there is no meaningful data in any of the rows, the problem might not have a practical solution.) Where the pivot row and pivot column converge is known as the pivot location.

6. Now, we use basic row operations to scale the pivot element to 1. Then utilizing this pivot one to set the value of all other elements in the pivot column to zero. A column that is entirely zero except for one is called a unit column. After that, we return to Step 4.

7. A variable that matches a unit column is known as a fundamental variable. Variables that are not basic are called non-basic variables. The item in the constant column in the row containing the one determines the value of each fundamental variable. Non-basic variables have no value at all. This is the best choice.

3. Presentation of Data

Assumption for the problem:

- The number of ingredients utilized overall is thought to be fixed.
- The variables included in this paper are predicted to have a linear relationship.

Table 3: A locally accessible ingredients is necessary for best formulation to produce one tones of floating feed for fingerling period of Siluriformes (catfish)

Ingredient	Inclusion level (%)	Quantity (in kg)	Cost (Taka)
Soybean meal	20%	200	10800
Meat and bone meal	30%	300	30000
Mustered oil cake	25%	250	10750
Salt	0.4%	4	76
Vitamin premix	9.6%	96	16320
Maize	15%	150	4500

Table 4: A locally accessible ingredients is necessary for best formulation to produce one tones of floating feed for grower's period of Siluriformes (catfish)

Ingredient	Inclusion level (%)	Quantity (in kg)	Cost(Taka)
Soybean meal	50%	500	27000
Meat and bone meal	30%	300	30000
Mustered oil cake	10%	100	4300
Salt	0.6%	6	114
Vitamin premix	9.4%	94	15980
Maize			

Table 5: A locally accessible ingredients to produce a tone each of both fingerling and grower of Siluriformes (catfish)

Ingredient	Maximum available (kg)	Cost (Taka)
Soybean meal	1000	54000
Meat and bone meal	1000	100000
Mustered oil cake	350	15050
Salt	10	190
Vitamin premix	250	42500
Maize	450	13500

Table 6: The quantity of finished ingredients to make feeds of Siluriformes (catfish) fingerling and growers fish

Ingredient	Quantity (in kg)	Quantity (in kg)	Maximum available (kg)
Soybean meal	200	500	1000
Meat and bone meal	300	300	1000
Mustered oil cake	250	100	350
Salt	6	4	10
Vitamin premix	94	96	250
Maize	150		450
Cost	72446 Taka	77394 Taka	

The data collected for this research is based mainly on both primary and secondary sources. The kinds of ingredients that went into making up the ration are those that are discovered through the market survey. Furthermore, interviews with the fishing officers were conducted regarding the methods and ratios used in mixing the materials.

4. Data Analysis

Let, the fingerling feeds $= x_1$

and the grower's feeds $= x_2$

4.1 Objective function

Minimize $Z = 72446x_1 + 77394x_2$

4.2 The constraints

For soybean meal: $200x_1 + 500x_2 \leq 1000$

For meat and bone meal: $300x_1 + 300x_2 \leq 1000$

For mustered oil cake: $250x_1 + 100x_2 = 350$

For salt: $4x_1 + 6x_2 = 10$

For vitamin premix: $94x_1 + 96x_2 \leq 250$

For maize: $150x_1 + 0x_2 \leq 450$

4.3 Non-negativity restrictions

$x_1, x_2 \geq 0$

The following is the simplex table that is used to analysis the result.

Table 7: Final Iteration by Simplex Method

Row	Basis	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	BFS
1	x_3	0	0	1	0	0.73	-95.45	0	0	300
2	x_4	0	0	0	1	-0.55	-40.91	0	0	400
3	x_1	1	0	0	0	0.01	-0.09	0	0	1.00
4	x_2	0	1	0	0	0.00	0.23	0	0	1.00
5	x_7	0	0	0	0	-0.16	-13.27	1	0	60.0
6	x_8	0	0	0	0	-0.82	13.64	0	1	300
7	Z	0	0	0	0	7739286.27	7728396.45	0	0	-149840.00

5. Discussion of Results

The model suggests that the data gathered yields the best outcome. This simply shows that x_1 (the fingerling feeds) and x_2 (the growing feeds) meaningfully to increase the value of the objective function of the linear programming model. Therefore, the cost is reduced when the objective function is 149840 Taka with the fingerling feeds, $x_1 = 1$ (tone) and the grower's feeds, $x_2 = 1$ (tone).

6. Conclusion

There are many applications where the linear programming technique is being used, in the fields of agriculture, animal husbandry, and livestock, and it is used to determine the components of fish feed for fish farmers in order to increase fish productivity. From analysis of the paper, cost of product was reduced to 149840 taka in the final iteration and it is noted that one of a ton fingerling feeds was produced while the production level for growers feeds one of a ton. The profit for fish farmers will grow overall if the expense of feeding is reduced to a minimum.

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